

	INDICATIVE INSPECTION AND TEST PLAN FOR CONTROL VALVE	Document no.: AdVENT-STD- QC-ITP-004, Rev-00
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INDICATIVE INSPECTION AND TEST PLAN FOR CONTROL VALVE


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			CHETAN	DUM	RBM
Date	Rev.	Description	Prepared By	Reviewed By	Approved By



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RECORD OF REVISIONS

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1.0 Scope:

This Inspection and test plan covers the minimum testing requirements of the Control Valve. The scope of inspection shall be as per PO/contractual requirements. The Sub-contractor/Manufacturer/Vendor/Supplier shall prepare stage wise Inspection and Testing Plan in detail complying the applicable specification, MR & DS requirements and shall submit for review/approval.

2.0 Reference documents:

1. PO/MR/PR/Standards referred there in/Job specifications/Approved documents.

3.0 Legends: -

H: Hold for Inspection, do not proceed, R: Review/Check/Verify documents, W: Witness

PO: Purchase order; WPS: Welding procedure specification; PQR: Procedure qualification record; NDE: Non-destructive examination; PT: Penetrant testing; MT: Magnetic particle testing; RT: Radiographic testing; TPIA: Third party inspection agency; MTC: Material Test Certificate; TC: Test Certificate

4.0 Notes: -

1. This is Indicative Inspection and Test Plan (IITP) only. Vendor shall prepare detailed Inspection and Test Plan based on Project Specifications and IITP.
2. In case of conflict between purchase specification, contract documents and ITP, conditions that are more stringent shall be applicable.
3. Type/Special Test Certificate offered for review shall not be more than 5 years old from the PO date, in absence of the same, Type test has to be performed on each rating from accredited test agency.
4. Acceptance norms for all the activities shall be as per PO/PR/Standards referred there in/Job specification/Approved documents.
5. Vendor shall carry out additional testing required, if any not specified above, as per PO/Technical specification.
6. Calibration status of measuring and monitoring instruments/machines to be verified by TPIA during inspection activities.
7. ITP shall be approved by LTHE.

5.0 Definitions: -

Hold Point - Fabrication shall be kept under hold until inspection has been performed or written clearance is obtained by vendor.

Witness Point - Advise of inspection, Fabrication may be continued if inspector is not present after invitation issued in due time.

Review Point - Review of Test Reports/Certificates/Inspection Reports

Approval Point - Approval Required for Documents

		INDICATIVE INSPECTION AND TEST PLAN FOR CONTROL VALVE				IITP no. :
Sr. No.	Activity Description	Reference Document	Acceptance Criteria	Record Format	EXTENT OF INSPECTION	In:
						Vendor
1)	Review & Approval of Detailed Inspection & Test Plan.	Approved Drawing, Datasheet & Spec.	Approved Drawing, Datasheet & Spec.	Approved ITP	100%	H
2)	Raw Material, Incoming Materials & In Process inspection -					
2.1	Casting & Forgings (Body, Bonnet, Disc, Stem, Body ring)					
2.1.1	Visual/Surface Quality, Marking, Dimensions, NDE.	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Inspection report, MTC	As per Vendor's Sampling Plan	H
2.1.2	Chemical & Mechanical properties	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Inspection report, MTC	All Heats	H
2.2	Accessories (Positioners, Converters, Solenoid valves, limit switches, regulators, etc.)					
2.2.1	Model no., Make, functional check	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Inspection report, MTC	100%	H
2.3	All casting materials -					
2.3.1	Heat Treatment as per specification	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	HT Charts, MTC	All Lots	H
2.4	Castings & Forgings for Austenitic SS Body & Bonnet					
2.4.1	IGC Test, Stabilization Heat treatment (as applicable)	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	HT Charts, MTC	Per Heat, Per Solution Annealed lot	H
2.5	All castings					
2.5.1	Radiography	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	RT Film / RT Report	AS per Data Sheets./ procedure	H
2.6	Bars for Trim Materials					
2.6.1	Chemical analysis	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Lab Report	All Heats	H
3)	Bought out Finished & Sub Contract Items					
3.1	Stems, Gaskets, Gear units, Fasteners, Glands, Packing, etc.					
3.1.1	Surface defects & Dimensions	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Inspection Report	As per vendor Sampling Plan	H

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3.1.2	Physical/Mechanical, Chemical Properties	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Material Test Certificate	All Heats	H
3.2	Machining of Components & Welding -					
3.2.1	Visual & Dimensional inspection, finishing, WPS/PQR	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Inspection Reports, Approvals, Certifications.	100%	H
3.2.2	PWHT (As applicable)	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	HT Charts, MTCs, Reports	100%	H
4)	Welding Inspection-					
4.1	Previous Qualifications for Overlay/Stelliting (Body seat ring/ Wedge)					
4.1.1	Overlay thickness, Stelliting, Hardness	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	WPS/PQR/ WPQ	100%	H
4.2	New qualifications for Overlay/Stelliting (Body seat ring/Wedge)					
4.2.1	Overlay thickness, Stelliting, Hardness	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	WPS/PQR/ WPQ	100%	H
4.3	NDE after Stelliting / Overlay (as applicable)					
4.3.1	Visual, PT test	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Test Reports	100%	H
4.4	NDE of welded joints-					
4.4.1	Radiography	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	RT films/ RT reports	100%	H
4.5	Post Weld Heat treatment (PWHT), as applicable-					
4.5.1	PWHT of welds	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Time-Temperature graph	100%	H

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Sr. No.	Activity Description	Reference Document	Acceptance Criteria	Record Format	EXTENT OF INSPECTION	Inspection	
						Inspected by	Vendor
5)	Product Evaluation & Testing-						
5.1	Finished Valve						
5.1.1	Shell Test: Gate/Globe/Check Back Seat: Gate/Globe Seat test: Gate/Globe/Check	Approved Drawing, Datasheet, procedure & Spec.	Approved Drawing, Datasheet & Spec.	Test Reports	100%	H	
5.1.2	Any mandatory test or Supplementary test, Helium leak test (as applicable)	Approved Drawing, Datasheet, procedure & Spec.	Approved Drawing, Datasheet & Spec.	Test Reports	100%	H	
5.1.3	Functional / Operational test (Motor/Actuator operated)	Approved Drawing, Datasheet, procedure & Spec.	Approved Drawing, Datasheet & Spec.	Test Reports	100%	H	
5.1.4	Test on positioners, Control valve dynamic response & response time verification; tests on accessories	Approved Drawing, Datasheet, procedure & Spec.	Approved Drawing, Datasheet & Spec.	Test Reports	100%	H	
5.1.5	Pneumatic test on actuator; Type test for valves in cryogenic service (if specified)	Approved Drawing, Datasheet, procedure & Spec.	Approved Drawing, Datasheet & Spec.	Test Reports	100%	H	
5.2	Holiday Test for coatings (as applicable)						
5.2.1	Defects/Pin holes/Cracks	Approved Drawing, Datasheet, procedure & Spec.	Approved Drawing, Datasheet & Spec.	Test Reports	100%	H	
6)	Final Inspection / Tests -						
6.1	Finished valve (Body, Bonnet, Stem)						
6.1.1	PMI check (for AS/SS)-	Approved Drawing, Datasheet & Spec.	Approved Drawing, Datasheet & Spec.	Inspection/Test Report	As per approved procedure	H	
6.1.2	Visual, marking, dimensions, bevel, special requirements if any, Painting of Carbon & Low alloy steel & Provision of bypass arrangement (as applicable)	Approved Drawing, Datasheet & Specifications.	Approved Drawing, Datasheet & Specifications.	Inspection Reports	100%	H	

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Sr. No.	Activity Description	Reference Document	Acceptance Criteria	Record Format	EXTENT OF INSPECTION	Inspection	
						Vendor	Material
6.2	Strip check, Differential hardness (as applicable)						
6.2.1	Verify components	Approved Drawing, Datasheet & Spec.	Approved Drawing, Datasheet & Spec.	Inspection/Test Report	1 Valve per Type/size & order	H	
6.3	Pre-qualification tests						
6.3.1	Fire safe test (soft seated ball, plug, butterfly & lubricated plug valve) Cryogenic test	Approved Drawing, Datasheet & Spec.	Approved Drawing, Datasheet & Spec.	Acceptance report	1 Valve per Type/size & order	H	
7)	Type Test -						
7.1	Type test / Prototype test certificate review.	Approved Drawing, Datasheet & Spec., Manufacturer Manual	Approved Drawing, Datasheet & Spec.; Manufacturer Standards	Inspection Report	Sample	H	
8)	MTC's & Certifications Review. (MTCs, Certificate from PESO/Statutory approval for use in specified hazardous area.	MTCs, Certificates, Test Reports, MR/PR, Specifications, Approved Drg,	MTCs, Certificates, Test Reports, MR/PR, Specifications, Approved Drg,	MTCs, Certificates, Test Reports. PESO, Statutory Approvals.	100%	H	
9)	Spares Quantity Check	Approved PR / Spare requirements, BOM.	Approved PR / Spare requirements	Packing List. BOM, List of Spares	100%	H	
10)	Painting & Packing - (Visual check & Suitable protection to prevent entry of foreign material)	Approved Drawing, Datasheet & Specifications.	Approved Drawing, Datasheet & Specifications.	Inspection Report	100%	H	

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Sr. No.	Activity Description	Reference Document	Acceptance Criteria	Record Format	EXTENT OF INSPECTION	In:
						Vendor
11)	Inspection Release Note.	FAT Reports, MTCs, Certificates Etc.	Approved Drawing, Datasheet & Specifications.	IRN	100%	-
12)	Submission of certificates. (final documentation including all inspection & material certificates)	PO, MR/PR, ITP, Approved Drawings / Approved Procedures / Specifications	PO, MR/PR, Approved Drawings / Approved Procedures / Specifications	Final Documents/Certificates, Test Report / MDR	100%	H